

Chapter 3: In-Class Exercises

part 2

3.6

Problem: Sasha is obligated to repay a loan made on the first of the month by paying \$80 at the end of each of the next thirty months, including this month. The effective monthly rate on the loan is .4%. Find Sasha's loan balance immediately after her twelfth payment.



$$i = .004$$

$$\begin{aligned} OLB(12) &= 80 a_{\overline{18}|.004} \\ &= 1386.71 \end{aligned}$$

3.6

Problem: A loan of \$20,000 is being repaid by payments of \$2,500 at the end of each year and a final smaller payment one year after the last \$2,500 payment. The annual effective interest rate on the loan is 8%. Find the outstanding loan balance just after the borrower has made payments totaling \$15,000.

$$\frac{15,000}{2,500} = 6$$

n isn't given!

$$\begin{aligned} OLB(6) &= 20,000 (1.08)^6 - 2500 s_{\overline{6}|.08} \\ &= 13397.66 \end{aligned}$$

3.6

Problem: Mr. and Mrs. Harper purchase a home for \$256,000. They make a down payment of \$40,000 and finance the remainder of the purchase price with a thirty-year mortgage at an annual effective interest rate of 6.5%. The Harpers sell their home at the end of eight years, just after having made their ninety-sixth end-of-month mortgage payment. The sales price is \$282,000, and the Harpers closing costs are 3% of the selling price. The outstanding loan balance is deducted from the amount the Harpers receive and sent to the lender. How large a check do the Harpers receive, and how much interest did they pay over the eight years?

$$256,000 - 40,000 = 216,000 = L \quad n = 30 \cdot 12 = 360$$

$$6.5\% = \text{yearly effective}$$

$$(1.065)^{1/12} - 1 = .005261694 = \text{monthly effective}$$

$$\text{Loan pmts} \Rightarrow 216,000 = Pa_{\overline{360}|.005261694}$$

$$P = 1338.96$$

$$\begin{aligned} \text{OLB}(96) &= 216,000 (1.065)^8 - 1338.96 S_{\overline{96}|.005261694} \\ &= 190,800.24 \end{aligned}$$

$$282,000 (.97) - \text{OLB}(96) = \boxed{82739.75}$$

$$\begin{aligned} \text{Interest paid} &: 1338.96(96) - (216,000 - 190,800.24) \\ &= \boxed{103,340.40} \end{aligned}$$

3.6

Problem: Waswate took out a sixty-month car loan with an interest rate of 3% convertible monthly. To repay the loan, Waswate was to make monthly payments of \$252.65. During the first three years of the loan, he made all of the payments as scheduled except that he neglected to pay the fourteenth and thirtieth. What is the outstanding balance at the end of three years, just after he has made the payment scheduled at that time?

$$n = 60 \quad i = \frac{.03}{12} = .0025 \quad \text{pmt} = 252.65$$



$$OLB(36) = 252.65 a_{\overline{24}|.0025}$$

$$+ 252.65 (1.0025)^{22}$$

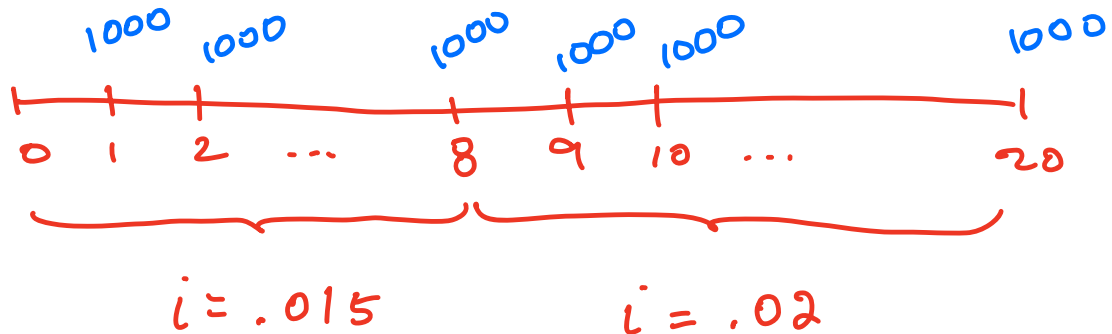
$$+ 252.65 (1.0025)^6$$

$$= 6401.53$$

3.6

Problem: Suppose that a loan is to be repaid by twenty end-of-quarter payments of \$1,000. The interest rate for the first two years is 6% convertible quarterly, and for the last three years it is 8% convertible quarterly. Find the loan amount, the outstanding loan balance just after the sixth payment, and the outstanding loan balance immediately after the fifteenth payment.

$$n = 20 \quad \text{pmt} = 1000$$



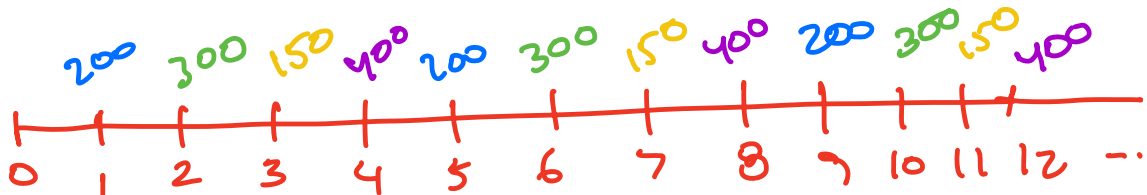
$$\begin{aligned}
 \text{OLB}(6) &= \text{time 6 value of PMTS 7, 8} \\
 &\quad + \text{time 6 value of PMTS 9-20} \\
 &= 1000 (1.015)^{-1} + 1000 (1.015)^{-2} \\
 &\quad + 1000 a_{\overline{12}|.02} (1.015)^{-2} = 12220.96
 \end{aligned}$$

$$\text{OLB}(15) = 1000 a_{\overline{5}|.02} = 4713.46$$

3.7

Problem: An annuity lasts for twelve calendar years. At the end of each quarter, there is a payment. First quarter payments are \$200, second quarter payments are \$300, third quarter payments are \$150, and fourth quarter payments are \$400. Find the accumulated value of this annuity just after the last payment using a nominal quarterly interest rate of 7.2%.

Not Covered on test 2.



Hint: write in terms of 4 different annuities.

3.8

Problem: On June 15, 1975, Roy purchased an annuity-immediate with annual payments for twenty-five years. The first payment was \$800 and the payments increased by 3% each year. The purchase price was based on an annual effective interest rate of 7%. Find this price.

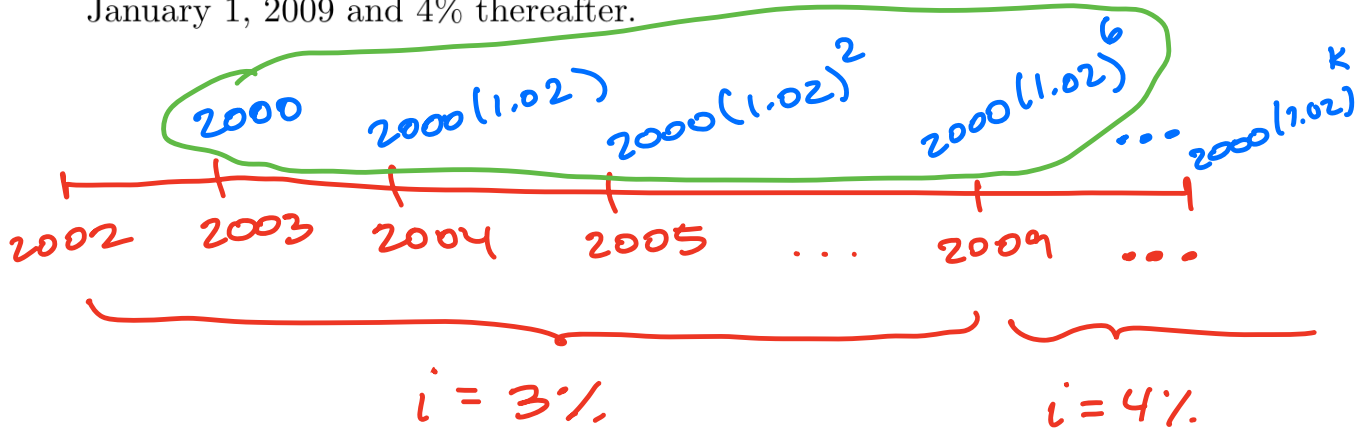
$$n = 25, \quad g = .03 \quad i = .07 \quad P = 800$$

$$PV = 800 \left(\frac{1 - \left(\frac{1.03}{1.07} \right)^{25}}{.07 - .03} \right)$$

$$= 12284.46$$

3.8

Problem: On January 1, 2002, Andrea inherits a perpetuity-immediate with annual payments. The first payment is \$2,000 and after that the payments increase by 2% each year. Find the value of this perpetuity on January 1, 2002 if the annual effective rate of interest is 3% from January 1, 2002 through January 1, 2009 and 4% thereafter.



$$PV = 2000 \left(\frac{1 - \left(\frac{1.02}{1.03} \right)^7}{.03 - .02} \right) \leftarrow \begin{array}{l} \text{First 7} \\ \text{pmts} \end{array} + (1.03)^{-7} \cdot 2000(1.02)^7 \left(\frac{1}{.04 - .02} \right) \begin{array}{l} \text{the} \\ \text{rest} \end{array}$$

$$= 106,601.34$$

3.9

Problem: Susan receives an annuity for her eighteenth birthday. The annuity pays \$2,000 on her nineteenth birthday and then has an annual payment on each of her birthdays through her thirty-fifth birthday. Each year the payments increase by \$500. Find the value of the annuity on Susan's eighteenth birthday, assuming the annual effective rate of interest is 4.2%.

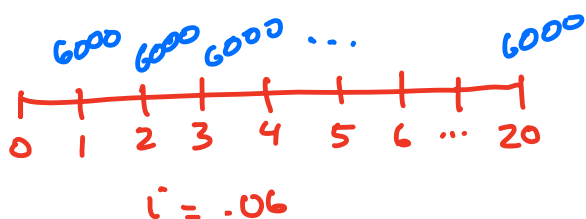
$$(I_{2000,500} a)_{\overline{17}.042}$$

$$= 2000a_{\overline{17}.042} + \frac{500}{.042} \left(a_{\overline{17}.042} - \frac{17}{1.042^{17}} \right)$$

$$= 66008.43$$

3.9

Problem: At the end of each year Mr. Dunn deposits \$6,000 into an investment fund. The fund pays out interest each year at an annual effective interest rate of 6%. Mr. Dunn is only able to reinvest this interest at an annual effective interest rate of 4%. Assuming Mr. Dunn continues to make his \$6,000 deposits for twenty years, that he always immediately deposits all the interest paid out into the 4% account, and that he makes no withdrawals, what is the accumulated value of Mr. Dunn's investments at the time of the last deposit?

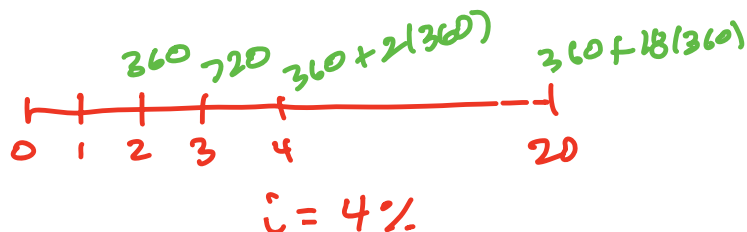


$$6000 \rightarrow 6000(1.06)$$

$$= 6360$$

$$2(600) \rightarrow 2(6000)(1.06)$$

$$= 12000 + 720$$

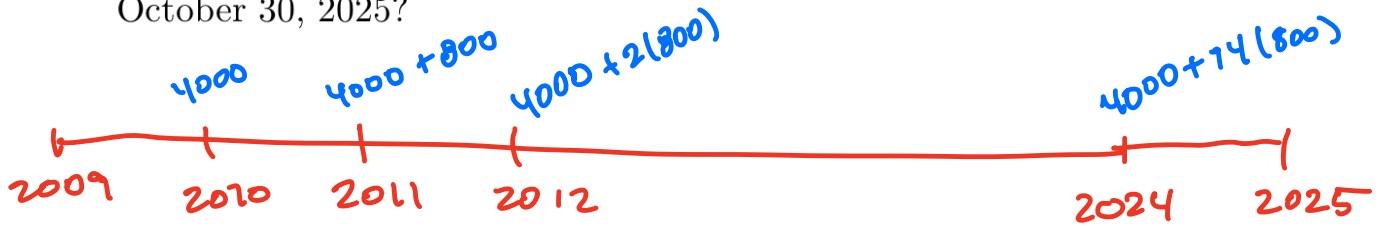


$$\text{Time 20} = 6000(20) + (I_{360,360} s)_{\overline{19}.04}$$

$$= 208002.71$$

3.9

Problem: Alfonso has an annuity that will pay \$4,000 on October 30, 2010 and has annual payments through October 30, 2024. The payments increase by \$800 each year. As soon as Alfonso receives a payment, he will deposit it in a savings account with a 4% annual effective interest rate, and he will not make any withdrawals. What will be the balance in Alfonso's 4% account on October 30, 2025?



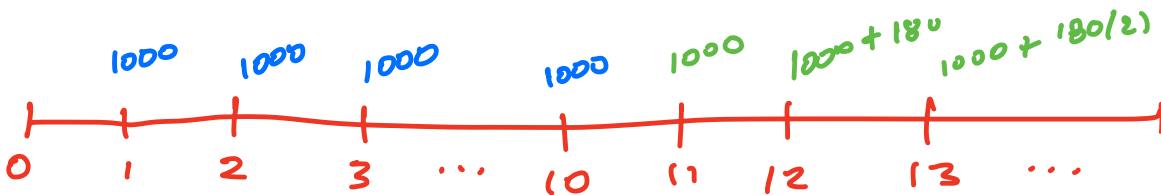
$$(I_{4000, 800} S)_{\overline{15}|.04} (1.04)$$

$$= \left(4000 S_{\overline{15}|.04} + \frac{800}{.04} (S_{\overline{15}|.04} - 15) \right) (1.04)$$

$$= 187788.75$$

3.9

Problem: Rafael purchases a perpetuity-immediate. The perpetuity pays \$1,000 at the end of each of the first eleven years and then has payments that increase by \$180 each year. So, the payment at time 11 is \$1,000, and the payment at time 12 is \$1,180. The purchase price was determined using a 5% annual effective interest rate. Find Rafael's purchase price.



$$i = 5\%$$

$$PV = 1000 a_{\overline{10}|1.05} + (1.05)^{-10} (I_{1000, 180})_{\overline{\infty}|.05}$$

$$= 64,201.75$$